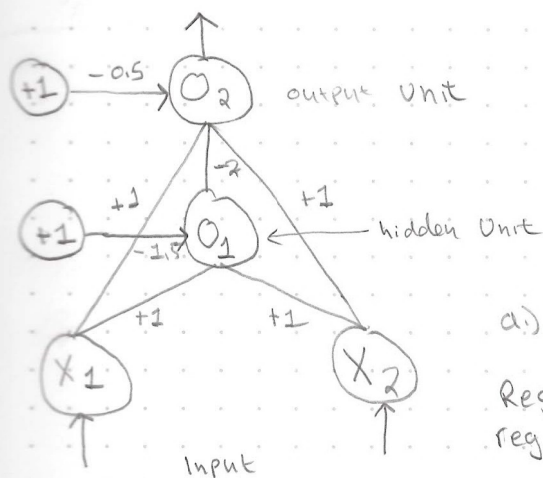


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Homework Assignment #2

Q.) Show that the below network solves XOR by constructing the decision regions for O_1 & O_2 and a truth table for the network

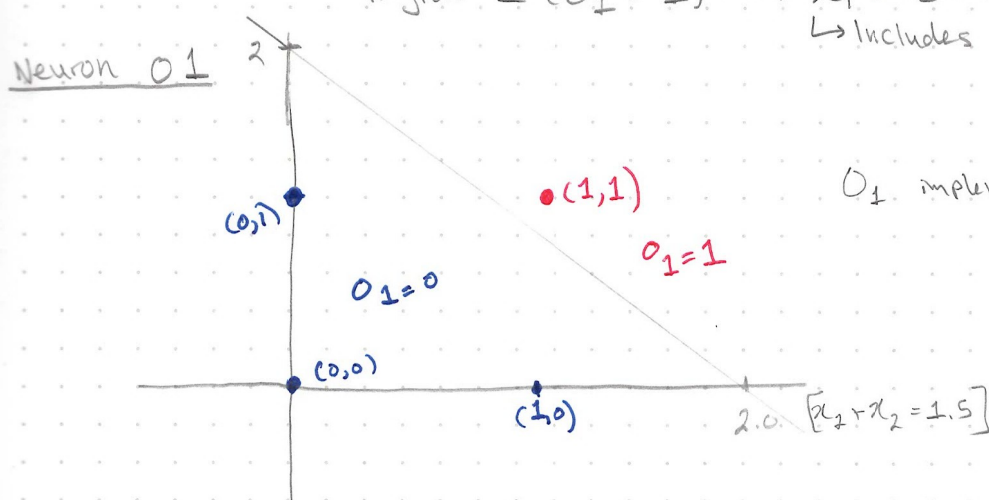


Input	Output
00	0
01	1
10	1
11	0

a) Decision regions for O_1 $\rightarrow x_1 + x_2 - 1.5 = 0$

Region 0 ($O_1 = 0$): Defined by $x_1 + x_2 < 1.5$. This region includes input points (0,0), (0,1), and (1,0)

Region 1 ($O_1 = 1$) $\rightarrow x_1 + x_2 \geq 1.5$
 $\hookrightarrow$ Includes input (1,1)



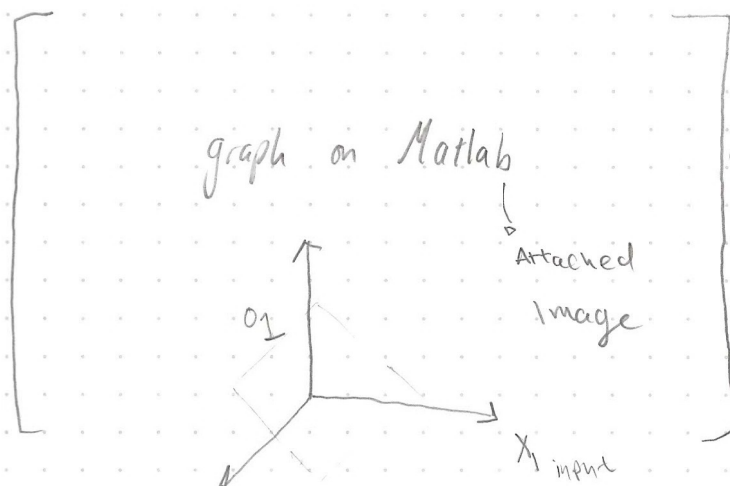
O_1 implements AND Function

b) Decision Regions for O_2 (Output neuron)

$\hookrightarrow$ receives 3 inputs $\rightarrow$ Decision Boundary is a plane in 3D with axes (x_1, x_2, O_1)

$$\rightarrow O_2 = \text{Step}(1 \cdot x_1 + 1 \cdot x_2 - 2O_1 - 0.5)$$

$$0 = x_1 + x_2 - 2O_1 - 0.5 \quad \leftarrow \text{Equation for decision Boundary plane}$$



Points above plane $\rightarrow O_2 = 1$

Points below plane $\rightarrow O_2 = 0$

For input (0,0,0) $\rightarrow 0+0-0-0.5 = -0.5$

For input (0,1,0) & (1,0,0) $\rightarrow$ sum is 0.5

For input (1,1,1) $\rightarrow 1+1-2(1)-0.5 = -0.5$